

$$n=5, \quad \downarrow 5$$

$$n=6, \quad \downarrow 21$$

limit = 500

```
for(int n=1; n<=limit; n++) { // O(limit)
```

```
    int sum=0;
```

```
    for(int i=1; i<=n; i++) { // O(N)
```

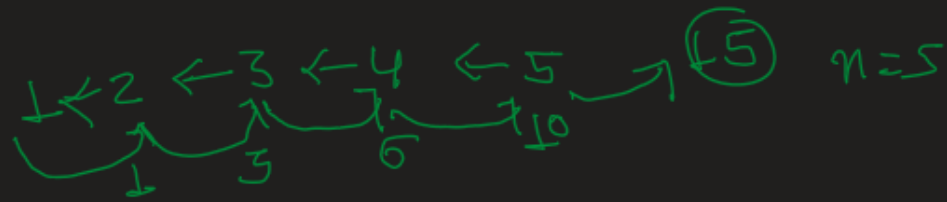
```
        sum += i;
```

```
    }
```

```
    cout << sum << "\n";
```

```
}
```

Recursive:-



```
int solveInRec(int n) {  
    if (n == 1)  
        return 1;
```

```
    int sum = n + solveInRec(n-1);
```

```
    return sum;  
}
```

```
for (int n=1; n<=limit; n++) // O(limit)
```

```
int sum = solveInRec(n); // O(N)
```

```
{  
    cout << sum << "\n";  
}
```

DP

```
graph TD; DP[DP] --> TD[Top Down]; DP --> BU[Bottom up]; TD --- M["(Memoization)"]; BU --- T["(Tabulation)"]
```

Top Down

(Memoization)

Bottom up

(Tabulation)

